SMDR: A Lightweight Detection-and-Repair Framework Against Temporal Misalignment Attacks on Multimodal 3D Detection for Autonomous Driving

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Abstract: 3D object detection is a core component of environmental perception in autonomous driving, with mainstream approaches fusing camera and LiDAR modalities to achieve more reliable perception results. Such multimodal 3D detectors assume that camera and LiDAR frames are temporally aligned. Recent research indicates that such detectors are vulnerable to temporal misalignment attacks. By merely injecting stale, frozen, or out-of-order frames into the sensor streams, without altering the model or the content of the frames themselves, the mAP of BEVFusion can be reduced from 0.685 to a range of 0.40–0.60. This paper proposes SMDR, a lightweight, unsupervised framework for detection and recovery that addresses seven types of temporal attacks. The detection module leverages three complementary forms of evidence: dynamic consistency, ego-pose consistency, and velocity parity. The repair module determines the appropriate recovery strategy based on whether frame information is missing: it either reorders frames or generates plausible replacement frames via extrapolation. Experiments were conducted on the BEVFusion model to simulate attacks and implement detection and repair mechanisms, using the nuScenes dataset for evaluation. Results show that the detection module achieved frame-level detection rates of 62.7%–95.7% against the most destructive LiDAR-side attacks (Constant: 90.2%, Random: 95.7%, Jitter: 87.5%, Drift: 71.5%, Scheduler: 62.7%, Burst: 67.4%, Reversal: 80.8%), with a false positive rate maintained between 5% and 14%. The repair module improves the mAP from 0.477 to 0.539 for the Constant/LiDAR setting and from 0.460 to 0.507 for the Reversal/LiDAR setting; it integrates seamlessly with BEVFusion without requiring modifications to the detector weights.

Keywords: Attack Detection, Autonomous Driving, Automotive Security

1. Introduction

Multi-sensor 3D perception enhances the reliability of object detection in autonomous driving by combining the geometric precision of LiDAR with the semantic richness of cameras. Bird's-Eye-View (BEV) fusion architectures further unify multi-modal features within the BEV space, establishing themselves as powerful backbones for 3D detection on benchmarks such as nuScenes [1]. However, recent studies on robustness benchmarks and sensor security [2], [3] indicate that multi-modal fusion does not offer complete protection; issues such as LiDAR damage, sensor failure, spatial and temporal misalignment, and single-modal attacks can significantly compromise fusion-based detectors.

One significant failure mode stems from temporal integrity. Recent studies indicate that attackers do not necessarily need to alter sensor data or model parameters; simply by delaying, replaying, or reordering genuine sensor frames, they can construct inputs that are temporally inconsistent yet physically plausible at the single-frame level [5]. In this scenario, while each individual image or point cloud frame may appear to be a valid observation, it no longer corresponds to the specific time step required for current detection and evaluation. Such mismatches are particularly hazardous for camera-LiDAR fusion detectors, as stale LiDAR geometric information can dominate the fused bird's-eye-view representation, causing the detector to output incorrect object states for the current moment.

However, in the context of in-vehicle perception, an alarm signal alone cannot directly recover the detection results. Applying a uniform repair method to all anomalous frames could fail to restore missing current observations or erroneously alter frames that still contain valid information, thereby further compromising detection performance. Consequently, repair strategies must first distinguish the source structure of the anomalous frames: where valid frames remain within the window but are merely out of sequence,

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output reordering maximizes the preservation of original observations; conversely, where current observations have been replaced by historical, frozen, or delayed frames, pose compensation based on ego-motion or extrapolation of the current representation from historical states is required.

At the same time, defense modules in autonomous driving systems must meet requirements for lightweight deployment. Retraining or fine-tuning the primary detector entails additional computational and engineering costs and relies on attack samples that are difficult to exhaustively enumerate. In contrast, runtime monitoring is better suited for deployment atop existing perception systems; it should leverage the detector's existing intermediate features, detection outputs, and ego-pose information, calibrating the normal dynamic range using clean sequences and performing anomaly detection during inference via low-dimensional descriptors and simple statistical tests.

To address this, we propose SMDR. This method leaves the backbone and detection head of BEVFusion intact, introducing only a lightweight detection-and-repair layer during inference. The detection component—calibrated using clean sequences—examines the temporal dynamics of BEV features, the spatial consistency between detection results and trusted ego-poses, and directional conflicts regarding object displacement and velocity (as estimated by the detection head) under time-reversal. The recovery component selects an operation based on the availability of source information within the stream: out-of-order frames undergo output reordering; stale LiDAR frames are corrected via ego-motion compensation; and any remaining alert frames that cannot be reliably recovered utilize BEV feature extrapolation as a fallback mechanism.

2. Related Work

2.1 Multimodal 3D Object Detection for Autonomous Driving

3D perception for autonomous driving typically relies on multiple sensors. Cameras provide dense semantic and textural cues, while LiDAR offers stable geometric and depth information; datasets such as nuScenes establish protocols for detection, tracking, and robustness evaluation based on continuous multi-sensor scenes [1]. The core challenge in multi-modal 3D detection lies in integrating observations—which differ in coordinate systems, densities, and noise characteristics—into a unified detection representation.

Camera-LiDAR fusion often occurs at the point level, proposal level, or near the detection head, typically requiring the projection of image features onto point clouds or candidate bounding boxes. TransFusion notes that hard projection-based association is fragile in the face of image degradation and calibration errors; consequently, it employs "soft association" at the query level to select relevant image information [6]. BEVFusion takes this further by unifying multi-sensor features within the Bird's-Eye-View (BEV) space, fusing camera semantics and LiDAR geometry into a shared BEV representation that supports both 3D detection and BEV map segmentation [4]. Meanwhile, BEVFormer demonstrates the importance of temporal BEV representations for 3D perception from a multi-camera visual perspective [7].

These methods operate on a key premise: the camera and LiDAR frames input at a specific inference moment describe the same physical instant in time. Temporal misalignment attacks specifically undermine this premise. The attack frame itself may consist of authentic sensor data, yet it originates from a different timestamp, while the detector is evaluated against the ego-pose and ground-truth annotations corresponding to the intended target moment. Consequently, the challenge shifts from designing new fusion backbones to monitoring and remediating existing BEV fusion detectors when temporal integrity is compromised.

2.2 Attack methods

Most perception attacks against autonomous driving systems target the data content itself. Representative attacks against LiDAR include placing physically realizable adversarial objects or physically removing points from real obstacles to hide targets or trigger false detections [8], [9]. Sensor fusion systems are not inherently secure either: the "frustum attack" bypasses simple cross-modal checks by maintaining semantic consistency between camera and LiDAR data, while "Fusion Is Not Enough" demonstrates that attacking a single modality can compromise a fusion-based detector [2], [3].

Temporal attacks operate in a different dimension. They require neither the fabrication of point clouds or image textures nor access to model weights; instead, the attacker simply ensures the detector receives a genuine frame—captured at the wrong time—at the target timestamp. Research such as DejaVu has shown that timestamp manipulation, network-induced latency, and the replay of historical messages are sufficient to disrupt multimodal perception tasks [5]. While such inputs may appear perfectly valid when viewed in isolation, the anomaly arises from the inconsistency between the source timestamp and the target timestamp.

In the experimental setup of this paper, the seven types of temporal remapping can be categorized into two groups based on their source structure. Constant, Random, Jitter, Drift, Burst, and Scheduler attacks cause the detector to receive historical or delayed frames, often excluding the current real-time frame from the input stream; in contrast, Reversal is a permutation attack where the real-time frame remains within the processing window but appears out of sequence (played in reverse). This distinction is not merely a taxonomic detail but a critical factor determining the appropriate remediation strategy: the former relies more heavily on geometric compensation or extrapolation, whereas the latter prioritizes restoring the original temporal relationships and reordering the output. Existing detection methods, such as AION, primarily address misalignment issues but have not yet extended this analysis of source struc-

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tural differences to the selection of closed-loop remediation strategies [10].

2.3 Detection method

Detection approaches can be broadly categorized into three types: physical consistency checks, sequential statistical detection, and cross-modal alignment detection specifically designed for temporal misalignment. 3D-TC2 detects spoofed targets in LiDAR data by verifying object motion consistency, while Shadow-Catcher utilizes LiDAR shadow geometry to check for plausible occlusion relationships [11], [12]. A common feature of these methods is that they do not modify the primary detector directly; instead, they validate the detection output against motion or geometric constraints—an approach similar to the runtime monitoring strategy proposed in this paper.

Sequential statistical detection offers another class of lightweight tools. Techniques such as Kalman filtering, set-membership filtering, CUSUM, and SPRT accumulate evidence of anomalies across continuous observations, converting frame-by-frame residuals into online alerts [13], [14], [15], [16].

AION advances the field by addressing runtime detection: acting as a plug-in defense module, it employs continuity-aware contrastive learning to acquire temporally smooth multimodal representations. It then uses dynamic time warping to track cross-modal alignment paths and generates a misalignment score based on path deviations, enabling the detection of camera-only, LiDAR-only, and joint temporal attacks [10].

While SMDR shares the same problem context—temporal misalignment detection as AION, its focus differs. AION concentrates on learning a representation sensitive to temporal misalignment; in contrast, SMDR utilizes clean-calibrated, interpretable runtime evidence derived from a fixed BEVFusion [4] detector and further explores how detection signals can inform subsequent recovery decisions. Consequently, this paper focuses not only on whether misalignment can be detected but also on whether the detection results are sufficient to support reliable output recovery.

3. Threat Model

This paper considers temporal sensor attacks during the deployment phase. An attacker positioned on the data path between the sensors and the multimodal detector can, by disrupting time synchronization, tampering with timestamps, inducing transmission delays, or replaying historical messages, cause the detector to receive a genuine sensor frame from a source frame $s(t)$ at the target frame t [5]. The attacker does not modify the content of images or point clouds, alter model weights, or rely on internal model parameters; consequently, the single-frame input may remain content-wise legitimate, with the anomaly stemming primarily from the temporal mismatch between the source and target timestamps.

We formalize the attack as a source-frame mapping $s(t)$.

Table 1 Temporal remapping attacks considered in this work.

Attack	Mechanism	Effect on the sensor stream
Constant	Freeze	Repeats one fixed frame throughout the attack window.
Random	Replacement	Replaces each target frame with a randomly selected historical frame.
Jitter	Stochastic delay	Delivers frames with randomly varying delays.
Reversal	Reordering	Delivers the frames in a window in reverse temporal order.
Burst	Intermittent freeze	Repeats a frame for short bursts separated by normal frames.
Drift	Cumulative delay	Gradually increases the delay as the attack window progresses.
Scheduler	Scheduling delay	Applies delays determined by a round-robin or priority scheduling rule.

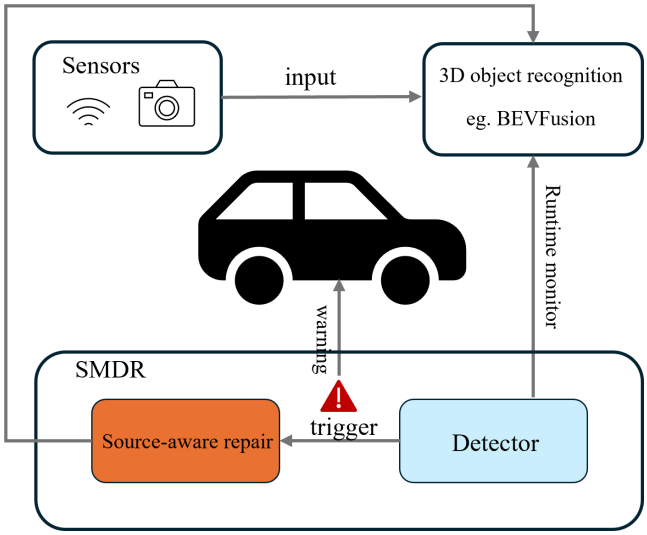


Fig. 1 Overview

If a frame is not attacked, then $s(t) = t$. Table 1 summarizes the seven temporal remapping patterns considered in this paper. The attacks can target the camera, LiDAR, or both simultaneously. We assume that vehicle localization, i.e., ego-pose, is provided by an independent trusted subsystem and lies outside the attack surface. Therefore, substituted sensor content may become inconsistent with the trusted ego-pose at the target frame.

The defender operates at the detector side, with access to input frames, BEV features, detection outputs, and the trusted ego-pose, making online, frame-by-frame decisions. The defender has no access to attack labels, true source indices, or adversarial samples; detection thresholds are calibrated solely using clean sequences. Our objective is to detect frames with temporal mismatches without modifying the primary detector's weights and to recover the detection output for the target timestamp as effectively as possible following an alert.

4. Proposed Method

Figure 1 illustrates the overall architecture of SMDR as a

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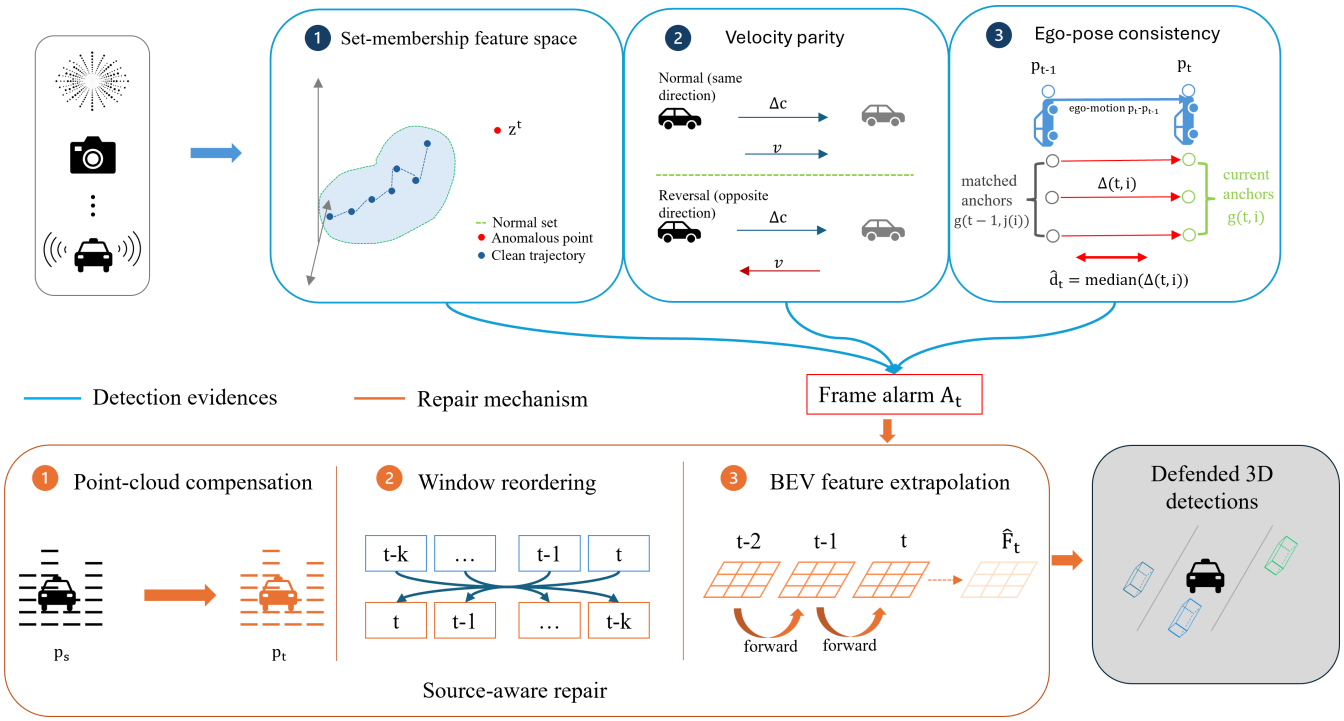


Fig. 2 Detail of SMDR.

lightweight plug-in for BEVFusion. The original BEVFusion perception pipeline remains unchanged: sensor frames are processed by the existing detector to generate 3D detection results. Concurrently, SMDR operates as a runtime monitor, reading runtime signals from the inference process, including fused BEV features, detection outputs, target velocities, and the trusted ego-pose. When the monitor triggers a frame-level alert, the source-aware repair module selectively replaces the affected outputs; otherwise, the original BEVFusion outputs are retained.

Figure 2 illustrates the detection and repair mechanism of SMDR. SMDR is a lightweight, unsupervised detection-and-repair framework calibrated using clean data. Instead of retraining BEVFusion, it utilizes runtime signals generated by BEVFusion during inference, including the fused BEV feature map, detection boxes, detection head regression velocities, and trusted ego-pose.

4.1 Detection side

The detection side incorporates three types of complementary evidence. The first type is set-membership feature monitoring, which determines whether the temporal variations in the fused BEV feature map remain within the normal dynamic range learned from clean sequences. The second type is velocity parity, which assesses whether the inter-frame displacement of an object aligns with the velocity regressed by the detection head under normal temporal progression. The third type is ego-pose consistency, which checks whether detected static content maintains spatial consistency with the current trusted ego-pose. These three types of evidence are fused via a logical OR operation to generate the final alert:

$$A_t = A_t^{\mathrm{smf}} \vee A_t^{\mathrm{par}} \vee A_t^{\mathrm{pose}}, \tag{1}$$

where A_t represents the final alert result for frame t.

4.1.1 Set-membership Feature Monitoring

This part asks a simple question: does the BEV representation of the current frame still lie on the normal temporal trajectory calibrated from clean sequences? Rather than judging whether an individual frame is real or fake, it checks whether the current representation can be reasonably predicted from the previous normal state. Historical, frozen, and delayed frames may remain individually legitimate while no longer following the feature evolution expected at the current time step.

Let t denote the frame index. At frame t, BEVFusion produces a BEV feature map $\boldsymbol{F}_t \in \mathbb{R}^{C\times H\times W}$, where C is the number of channels and H and W are the spatial dimensions. We apply global pooling and then PCA fitted on clean sequences to obtain a low-dimensional observation descriptor:

$$\boldsymbol{z}_t = \mathrm{PCA}_d(\mathrm{Pool}(\boldsymbol{F}_t)). \tag{2}$$

Here, $\mathrm{Pool}(\cdot)$ aggregates the feature map into a channel vector, $\mathrm{PCA}_d(\cdot)$ retains d principal components, and $\boldsymbol{z}_t \in \mathbb{R}^d$ is the resulting descriptor. We use $d = 32$.

We assume that clean descriptors evolve smoothly and model them using a first-order set-membership filter:

$$\boldsymbol{x}_t = \boldsymbol{A}\boldsymbol{x}_{t-1} + \boldsymbol{w}_t, \qquad \boldsymbol{z}_t = \boldsymbol{x}_t + \boldsymbol{v}_t. \tag{3}$$

Here, $\boldsymbol{x}_t$ is the unobserved normal state, $\boldsymbol{w}_t$ represents normal scene changes, and $\boldsymbol{v}_t$ represents small fluctuations introduced during feature extraction. The transition matrix is $\boldsymbol{A} = \alpha\boldsymbol{I}$, where $\alpha = 0.98$ and $\boldsymbol{I}$ is the identity matrix.

Because $\boldsymbol{x}_t$ is unavailable during calibration, SMDR uses $\boldsymbol{z}_{t-1}$ as a proxy for the preceding state and computes the clean one-step prediction residual:

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$$\boldsymbol{r}_t = \boldsymbol{z}_t - \boldsymbol{A}\boldsymbol{z}_{t-1}. \tag{4}$$

These residuals describe the temporal changes allowed in clean sequences and are used to calibrate the normal set.

During runtime, SMDR predicts the current normal state and measures the deviation of the observed descriptor:

$$\hat{\boldsymbol{x}}_{t|t-1} = \boldsymbol{A}\hat{\boldsymbol{x}}_{t-1|t-1}, \qquad \boldsymbol{e}_t = \boldsymbol{z}_t - \hat{\boldsymbol{x}}_{t|t-1}. \tag{5}$$

Here, $\hat{\boldsymbol{x}}_{t|t-1}$ is the state predicted for frame t using information available through frame $t-1$, and $\boldsymbol{e}_t$ is the prediction residual. SMDR estimates an uncertainty matrix $\boldsymbol{S}_t$ from clean residuals and computes

$$d_t^2 = \boldsymbol{e}_t^\top \boldsymbol{S}_t^{-1} \boldsymbol{e}_t. \tag{6}$$

Here, d_t^2 is the squared Mahalanobis distance. $\boldsymbol{S}_t^{-1}$ normalizes residuals according to the normal variation observed in different directions of the clean data. A larger d_t^2 therefore indicates a larger deviation from the normal temporal set.

Finally, SMDR applies SPRT so that a single noisy frame does not immediately trigger an alert:

$$\Lambda_t = \Lambda_{t-1} + \log \frac{p_1(d_t^2)}{p_0(d_t^2)}. \tag{7}$$

Here, Λ_t is the accumulated abnormal evidence up to frame t, p_0 is the normal distance distribution fitted on clean data, and p_1 is an enlarged abnormal distribution constructed from p_0 without attack samples. The channel raises an alert when the evidence exceeds threshold B_{sprt}:

$$A_t^{\mathrm{smf}} = \mathbb{I}\left[\Lambda_t \geq B_{\mathrm{sprt}}\right], \tag{8}$$

where $\mathbb{I}[\cdot]$ is the indicator function and $A_t^{\mathrm{smf}} = 1$ denotes an alert from the set-membership feature monitoring channel.

4.1.2 Velocity Parity

Set-membership feature monitoring alone may be insufficient for detecting Reversal attacks, because a reversed window still consists of temporally adjacent real frames whose feature descriptors and adjacent changes can remain within the normal set calibrated from clean data. SMDR therefore further seeks evidence of temporal reversal from object-level motion.

Under the normal temporal order, the displacement of a moving vehicle between two adjacent frames should agree with the velocity predicted by the detection head. If the input window is delivered in reverse order, the apparent displacement in the displayed sequence changes direction. However, the velocity predicted from each individual frame still reflects the motion direction at its original capture time. This creates a directional contradiction.

Let $\boldsymbol{c}_{t,i}$ denote the ground-plane center of object i at frame t, and let $\boldsymbol{v}_{t,i}$ denote its velocity predicted by the detection head. We first associate objects between adjacent frames under a class constraint and compute the inter-frame displacement:

$$\Delta\boldsymbol{c}_{t,i} = \boldsymbol{c}_{t,i} - \boldsymbol{c}_{t-1,j(i)}, \tag{9}$$

where $j(i)$ denotes the object in the previous frame matched to object i. To reduce noise from static objects and unreliable associations, we retain only objects with sufficiently large displacement and velocity, while excluding static classes such as barrier and traffic cone.

Then compares the displacement and velocity directions of each matched object and computes a speed-weighted frame-level parity score:

$$S_t^{\mathrm{par}} = -\frac{\sum_{i\in\mathcal{M}_t} \|\boldsymbol{v}_{t,i}\| \cos(\Delta\boldsymbol{c}_{t,i}, \boldsymbol{v}_{t,i})}{\sum_{i\in\mathcal{M}_t} \|\boldsymbol{v}_{t,i}\|}, \tag{10}$$

where $\mathcal{M}_t$ is the set of matched objects that pass the motion filter. Under normal temporal order, displacement and velocity point in similar directions, resulting in a positive cosine value and a low S_t^{par}. Under Reversal, they tend to point in opposite directions, making the cosine value negative and increasing the score.

Finally, SMDR calibrates the threshold τ_{par} as a high quantile of S_t^{par} on clean sequences:

$$A_t^{\mathrm{par}} = \mathbb{I}\left[S_t^{\mathrm{par}} > \tau_{\mathrm{par}}\right], \tag{11}$$

where $\mathbb{I}[\cdot]$ is the indicator function. When $A_t^{\mathrm{par}} = 1$, the velocity-parity channel flags frame t as potentially reversed.

4.1.3 Ego-pose Consistency

Ego-pose consistency uses the trusted ego-pose to check whether low-speed objects remain spatially consistent. Under normal operation, matched static anchors should remain nearly fixed after detections are transformed into global coordinates. When historical or reordered content is interpreted using the current ego-pose, multiple anchors exhibit a shift in a similar direction.

SMDR selects low-speed detections as anchors and matches them across adjacent frames under a class constraint. Let $\boldsymbol{g}_{t,i}$ be the global center of anchor i at frame t. Its displacement from the matched anchor $j(i)$ in the previous frame is

$$\boldsymbol{\delta}_{t,i} = \boldsymbol{g}_{t,i} - \boldsymbol{g}_{t-1,j(i)}, \qquad \hat{\boldsymbol{d}}_t = \mathrm{median}_{i\in\mathcal{A}_t} \boldsymbol{\delta}_{t,i}, \tag{12}$$

where $\mathcal{A}_t$ is the set of successfully matched anchors and $\hat{\boldsymbol{d}}_t$ represents their shared displacement.

SMDR scores this displacement by its magnitude, its projection along ego motion, and the unmatched-anchor ratio:

$$\begin{aligned} S_t^{\mathrm{mag}} &= \left\|\hat{\boldsymbol{d}}_t\right\|, \\ S_t^{\mathrm{proj}} &= \frac{\hat{\boldsymbol{d}}_t^\top \boldsymbol{u}_t}{\|\boldsymbol{p}_t - \boldsymbol{p}_{t-1}\|}, \qquad \boldsymbol{u}_t = \frac{\boldsymbol{p}_t - \boldsymbol{p}_{t-1}}{\|\boldsymbol{p}_t - \boldsymbol{p}_{t-1}\|}, \\ S_t^{\mathrm{unm}} &= 1 - \frac{|\mathcal{A}_t|}{|\mathcal{R}_{t-1}|}. \end{aligned} \tag{13}$$

Here, $\boldsymbol{p}_t$ is the translation of the trusted ego-pose, $\boldsymbol{u}_t$ is the ego-motion direction, and $\mathcal{R}_{t-1}$ is the previous anchor set. The projection term is omitted when ego motion is too small.

The clean-calibrated thresholds are fused into the final score:

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$$S_t^{\text{pose}} = \max\left(\frac{S_t^{\text{mag}}}{\tau_{\text{mag}}}, \frac{S_t^{\text{proj}}}{\tau_{\text{proj}}}, \frac{S_t^{\text{unm}}}{\tau_{\text{unm}}}\right) \tag{14}$$

$$A_t^{\text{pose}} = \mathbb{I}\left[S_t^{\text{pose}} > \tau_{\text{pose}}\right]. \tag{15}$$

All thresholds are calibrated from clean data, and $A_t^{\text{pose}} = 1$ denotes an ego-pose consistency alert.

4.2 Source-aware Repair

When the detector raises a frame-level alert $A_t = 1$ at frame t, SMDR selects a repair action according to whether the correct observation remains in the stream. Under Reversal, real observations are still available but delivered in the wrong order, so SMDR reorders existing outputs. Under historical-frame, frozen-frame, and delay attacks, the current observation is replaced by stale content; SMDR first attempts point-cloud compensation and otherwise predicts the current BEV feature map from trusted history.

4.2.1 Point-cloud Compensation.

For an alerted LiDAR frame, SMDR matches the current low-speed anchors to a trusted bank from the most recent unflagged frame and estimates their shared displacement $\hat{\boldsymbol{d}}_t^{\text{src}}$. The superscript src denotes a repair-stage content–pose shift and distinguishes it from the adjacent-frame displacement used by the detection channel. SMDR searches the trusted ego-pose history for the most likely source frame:

$$\hat{s}_t = \arg\min_{s<t} \left\| (\boldsymbol{p}_t - \boldsymbol{p}_s) - \hat{\boldsymbol{d}}_t^{\text{src}} \right\|. \tag{16}$$

Here, $\hat{s}_t$ is the estimated source-frame index, and $\boldsymbol{p}_t$ and $\boldsymbol{p}_s$ are the trusted ego-pose translations at the current frame and historical candidate frame, respectively. If the minimum residual is too large, SMDR retains the raw output. Otherwise, it compensates the received point cloud $\boldsymbol{P}_t$ from the estimated source pose to the current pose:

$$\tilde{\boldsymbol{P}}_t = T_{L_{\hat{s}_t} \to L_t} \boldsymbol{P}_t, \qquad T_{L_{\hat{s}_t} \to L_t} = T_{G \to L_t} T_{L_{\hat{s}_t} \to G}. \tag{17}$$

Here, $L_{\hat{s}_t}$, L_t, and G denote the estimated source LiDAR frame, the current LiDAR frame, and the global coordinate system. SMDR re-runs BEVFusion using $\tilde{\boldsymbol{P}}_t$.

4.2.2 Reversal Window Reordering.

For a Reversal window supported by velocity parity, real observations are not lost. SMDR therefore reorders the raw detection outputs instead of generating new boxes. If the source mapping inside the window follows

$$s(t) = c - t, \tag{18}$$

where $s(t)$ is the source-frame index of target frame t and c is the reversal axis, each flagged target frame q with an estimated source index $\hat{s}_q$ votes for

$$c_q = q + \hat{s}_q. \tag{19}$$

SMDR takes the median vote in the window $\mathcal{W}$, $\hat{c} = \text{median}_{q \in \mathcal{W}} c_q$, and restores the output by

$$\hat{\boldsymbol{y}}_t = \boldsymbol{y}_{\hat{c}-t}, \tag{20}$$

where $\boldsymbol{y}_t$ and $\hat{\boldsymbol{y}}_t$ are the raw and reordered detection outputs, respectively.

4.2.3 BEV Feature Extrapolation.

If point-cloud compensation or window reordering is unreliable, SMDR uses the most recent $K = 3$ trusted (unflagged) BEV feature maps as history. It first aligns them to the current coordinate system using ego motion and then predicts the current feature map:

$$\hat{\boldsymbol{F}}_t = \tilde{\boldsymbol{F}}_{t-1} + g_\phi\left(\tilde{\boldsymbol{F}}_{t-K:t-1}, \boldsymbol{e}_t\right). \tag{21}$$

Here, $\tilde{\boldsymbol{F}}_{t-K:t-1}$ denotes the aligned historical feature maps, $\tilde{\boldsymbol{F}}_{t-1}$ is the most recent map, $\boldsymbol{e}_t$ contains the relative ego-motion conditions, and g_ϕ is an extrapolation network trained only on clean sequences. The detection head directly decodes the final result from the predicted feature map $\hat{\boldsymbol{F}}_t$.

5. Experimental

5.1 Experimental Setup

We evaluate SMDR on the nuScenes validation set [1], which comprises 6,019 annotated frames. BEVFusion [4] is employed as the primary detector. All experiments utilize official pre-trained weights; the primary detector is neither fine-tuned nor modified in terms of model parameters. With clean inputs, BEVFusion achieves an mAP of 0.685 and an NDS of 0.713.

We evaluate seven types of temporal remapping attacks: Constant, Random, Reversal, Jitter, Drift, Burst, and Scheduler. Each attack is applied to three modalities, camera-only, LiDAR-only, and both, resulting in 21 configurations. The attack window initiates every 25 frames and lasts for 10 frames, while the first 25 frames of each scene remain clean.

All calibration is performed using only the first 100 attack-free nuScenes scenes. Each scene is an ordered sequence of frames. The PCA projection and the thresholds for set-membership feature monitoring, velocity parity, and ego-pose consistency are fitted from the frames in these scenes while preserving their temporal order and scene boundaries. Neither adversarial samples, attack labels, nor ground-truth source frame indices are used during calibration or online defense. The impact of attacks is assessed using mAP and NDS; frame-level detection is evaluated via TPR/FPR; and closed-loop recovery is measured using post-defense mAP and the recovery rate:

$$\text{Recovery} = \frac{\text{mAP}_{def} - \text{mAP}_{atk}}{\text{mAP}_{clean} - \text{mAP}_{atk}} \tag{22}$$

5.2 Attack Configurations and Baseline Performance

Existing studies [5], [10] have systematically demonstrated the disruptive impact of temporal misalignment attacks on multimodal perception. We replicate the experimental setup using BEVFusion and the nuScenes validation set to establish a baseline for evaluating subsequent detection and mitigation methods.

Table 2 reports the impact of temporal attacks on the vanilla BEVFusion baseline under 21 attack configurations;

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no SMDR detection or repair is enabled in this table. The results align with existing findings: camera-only attacks have a minimal impact on BEVFusion, with mAP remaining between 0.663 and 0.679; LiDAR-only attacks cause a significant performance drop, with mAP falling to the 0.437–0.603 range; and both attacks result in a further decline to 0.398–0.590. Consequently, the remainder of this paper focuses on analyzing LiDAR-side attacks and the high-impact configurations that require mitigation.

Table 2 Impact of temporal attacks on the vanilla BEVFusion.

Attack	Camera	LiDAR	Both
Constant	0.668 / 0.704	0.477 / 0.603	0.443 / 0.586
Random	0.663 / 0.702	0.437 / 0.584	0.398 / 0.564
Reversal	0.666 / 0.703	0.460 / 0.594	0.428 / 0.578
Jitter	0.676 / 0.708	0.556 / 0.644	0.536 / 0.634
Drift	0.673 / 0.706	0.515 / 0.622	0.488 / 0.609
Burst	0.679 / 0.710	0.603 / 0.669	0.590 / 0.662
Scheduler	0.676 / 0.708	0.545 / 0.638	0.525 / 0.628

5.3 Detection results

Table 3 presents the frame-level detection results of SMDR against seven types of temporal attacks. For LiDAR-side attacks, SMDR achieves a True Positive Rate (TPR) of 62.7%–95.7%. Among these, the Constant and Random attacks are the easiest to detect, as repetitive or stale point clouds significantly disrupt BEV feature dynamics; the "Reversal" attack, meanwhile, relies on supplementary evidence regarding velocity parity and ego-pose consistency, achieving TPRs of 80.8% and 86.5% in the LiDAR-only and both-modality settings, respectively.

In contrast, the TPR for camera-only attacks ranges from only 10.3% to 21.1%. This indicates that within BEVFusion, camera-specific temporal mismatches have a weaker impact on final 3D detection, and anomalous signals are more difficult to isolate from the fused features. Detection results for the both-modality setting are generally similar to those of the LiDAR-only setting, suggesting that the primary observable anomalies in both-modality attacks still originate from the LiDAR geometric flow.

Table 3 Frame-level detection performance of SMDR. Values are raw TPR/FPR in percent.

Attack	Camera	LiDAR	Both
Constant	18.1 / 8.4	90.2 / 5.2	90.7 / 11.1
Random	21.1 / 8.6	95.7 / 5.8	98.3 / 11.2
Reversal	10.3 / 8.5	80.8 / 10.5	86.5 / 10.8
Jitter	20.8 / 8.4	87.5 / 13.2	89.2 / 13.7
Drift	12.9 / 8.2	71.5 / 10.2	72.4 / 10.4
Burst	12.8 / 8.5	67.4 / 3.7	69.3 / 12.7
Scheduler	12.4 / 8.2	62.7 / 10.1	63.9 / 10.3

5.4 Repair results in Lidar side

Table 4 presents the repair results achieved by SMDR against LiDAR-side attacks. In the online setting, SMDR yields positive mAP gains across all seven attack categories, demonstrating that source-aware repair—triggered by an alert—can recover a portion of the perception performance.

Table 4 Repair performance of SMDR on LiDAR attacks. Values are mAP on nuScenes validation.

Attack	Attacked	Repaired	Δ	Recovery
Online source-aware repair				
Constant	0.477	0.539	+0.063	30.0%
Random	0.437	0.460	+0.022	9.0%
Reversal	0.460	0.507	+0.046	20.6%
Jitter	0.556	0.587	+0.031	24.1%
Drift	0.515	0.536	+0.021	12.5%
Burst	0.603	0.629	+0.026	32.1%
Scheduler	0.545	0.565	+0.020	14.4%
Offline oracle-source reference				
Constant	0.477	0.597	+0.120	57.8%
Random	0.437	0.560	+0.123	49.6%
Reversal	0.460	0.666	+0.206	91.7%
Jitter	0.556	0.644	+0.088	68.5%
Drift	0.515	0.626	+0.111	65.4%
Burst	0.603	0.660	+0.058	70.2%
Scheduler	0.545	0.640	+0.094	67.6%

Specifically, the mAP for the Constant attack rises from 0.477 to 0.539, representing a recovery rate of 30.0%; the Burst attack achieves a recovery rate of 32.1%, though the absolute gain is relatively modest due to the high baseline performance following the attack.

There are notable differences in repair gains across the various attack types. Random, Drift, and Scheduler exhibit lower online recovery rates, indicating that online source estimation remains the primary limiting factor when source frame relationships are unstable or latency patterns shift in a piecewise manner. Reversal differs from stale-frame attacks; since the true observations remain within the processing window, a recovery rate of 20.6% is achieved by combining window reordering with BEV extrapolation.

The offline oracle results in Table 4 serve only as a diagnostic reference. The repair router is given the ground-truth source mapping $s(t)$ recorded by the attack injector, but not clean target-frame observations. The known source relation is used to inverse-reorder available outputs for Reversal or to select the appropriate repair path for stale-frame attacks. Under oracle conditions, the recovery rates for Random, Jitter, Drift, and Scheduler are substantially higher than those in the online setting, suggesting that the main bottlenecks lie in online source estimation and repair routing rather than the repair operations themselves. For Reversal, the recovery rate reaches 91.7% with oracle source reordering, indicating that once the source order is correctly restored, this attack can be largely mitigated through reordering. The remaining gap to clean performance reflects sensor content that is missing or cannot be fully aligned after repair.

6. Limitation and Conclusion

Analysis of nuScenes sequences reveals a prevalence of scenarios characterized by stationary states, low-speed driving, and minimal object motion. This data characteristic significantly complicates the detection and mitigation of temporal attacks. In low-dynamic scenarios, geometric discrepancies between the current frame and historical, delayed, or reversed-window frames are subtle; consequently, relying solely on frame-to-frame feature residuals makes it difficult to reliably distinguish between normal and misaligned

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frames. While SMDR supplements detection evidence by incorporating object velocity parity and ego-pose consistency, "delay" and "reversal" attacks remain highly stealthy in scenarios lacking trackable moving objects.

Unlike AION, which employs contrastive learning and Dynamic Time Warping (DTW) for detection, SMDR prioritizes lightweight runtime monitoring. Instead of training an attack classifier, the detection module calibrates a set of normal dynamics using clean sequences and triggers alerts based on a combination of set-membership filtering, velocity parity, and ego-pose consistency. This design reduces deployment complexity and better aligns with the computational constraints for online detection in autonomous driving systems.

Limitations persist regarding the repair component. For attacks where the source frame remains within the temporal window, reordering preserves genuine observations; however, in cases involving historical frames, frozen frames, or missing current observations, BEV extrapolation relies on historical context to predict the current representation. The effectiveness of this approach is influenced by factors such as the number of objects, the magnitude of object motion, and changes in occlusion. When the current frame contains numerous dynamic objects or newly appearing ones, the extrapolated results may deviate from actual observations. Future work should involve visualizing and comparing predicted BEV, ground-truth BEV, and final detection boxes to analyze how extrapolation errors propagate to detection outcomes.

Furthermore, a limited drop in average mAP does not necessarily imply a weak attack. A more plausible explanation is that a significant portion of the dataset consists of stationary or low-motion scenarios, where 3D detection results are relatively insensitive to brief misalignments. In other words, average mAP metrics may mask the risks posed by temporal attacks in highly dynamic or safety-critical scenarios. Consequently, future evaluations should incorporate metrics such as dynamic scenario categorization, risks associated with nearby objects, and the status of the forward driving path to more precisely assess the impact of temporal attacks on autonomous driving perception safety.

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